

Treatment and Emergency Protocol for Maple Sugar Urine Disease (MSUD)

Patient:

DOB:

MBR#:

This person has MSUD and is at risk for serious complications from this condition.

Definition: MSUD is a rare inherited disorder of branched chain amino acid metabolism. A single enzyme (branched chain keto acid dehydrogenase) is missing and the essential amino acids VALINE, ISOLEUCINE AND LEUCINE cannot be metabolized past the first degradative step. The three ketacid derivatives are toxic metabolites and the reaction also reverses and the three amino acids – especially leucine-reach toxic levels.

Maintenance: Patients with MSUD are treated with special formula containing essential amino acids EXCEPT FOR THE THREE BRANCHED AMINO ACIDS and a LOW PROTEIN DIET to provide minimal essential VALINE, ISOLEUCINE AND LEUCINE.

Treatment For Metabolic Decompensation: A metabolic crisis may be triggered by a viremia, gastroenteritis or a problem with the diet, LETHARGY, CONFUSION, VOMITING, SEIZURES, KETOACIDOSIS, COMA are complications of MSUD.

Immediate and aggressive treatment is necessary.

1. Establish an IV and give 10% glucose with ½ normal saline at 1½ maintenance.
2. Aim to keep the blood glucose level above 120 mg/dl
If the glucose is over 180 you may use insulin at a low dose.
3. Add Arginine HCl (if available for injection) at 200 mg/kg to run over an hour then repeat every 8 hours.
4. If the patient is responding poorly but not vomiting place an NG tube for feeding with the patient's MSUD formula. (AIM: Adding other required amino acids to stimulate protein synthesis to use up Leucine which is the most neurotoxic metabolite).

Notification:

Please notify the Metabolic Specialist immediately at 859-323-0396 during working hours.

(After business hours, nights and weekends
call UK MD's and ask for Geneticist on call):

(After Hours: Please call 1-800-888-5533 or 859-257-5522)

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